

Lecture 31 — Live Kernel Patching

Jeff Zarnett

jzarnett@uwaterloo.ca

Department of Electrical and Computer Engineering
University of Waterloo

May 30, 2026

This lecture is not yet written. The outline lives in the plan file.

Intended sections:

- 1 What's hard about hot-patching
- 2 ftrace as the patching backbone
- 3 The historical pair — kpatch vs. kGraft
- 4 The consistency model
- 5 Shadow variables
- 6 What live patching can't do
- 7 Tooling
- 8 Live patching as fail-soft